

Degradation of Histamine Solutions Used for Bronchoprovocation*

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Study objectives: To determine optimal storage conditions for histamine diphosphate (HDP) solutions used for bronchoprovocation.

Design: HDP was dissolved in buffered saline solution to concentrations of 0.125 to 16 mg/mL and stored in 3-mL unit dose syringes at different temperatures for varying lengths of time, with and without protection from fluorescent light.

Setting: Dark freezer (-20°C), dark refrigerator (4°C), and laboratory counter top (20°C) illuminated by fluorescent light (375 foot-candles).

Measurements: HDP concentrations were measured after the solutions were prepared and during storage by a high-performance liquid chromatographic assay that differentiates histamine from its break down products.

Results: All dilutions were sterile after preparation and contained 97 to 110% of the labeled amount of HDP. Solutions constantly exposed to fluorescent light (375 foot-candles) and room temperature (20°C) contained only 20 to 37% of the initial concentrations after 7 days. The same dilutions stored at room temperature, but protected from light, contained 83 to 94% of the initial concentrations. Dilutions stored in the dark in a refrigerator (4°C) retained 97% of the initial concentrations after 8 weeks, while dilutions stored in the dark freezer (-20°C) were stable for 12 months.

Conclusions: Exposure to fluorescent light at room temperature results in degradation of histamine solutions used for bronchoprovocation. Dilutions stored in unit dose syringes and protected from light are stable for at least 8 weeks in the refrigerator and up to 12 months frozen. Once removed from the refrigerator or freezer, the solutions should be used within 6 h or discarded. (CHEST 1999; 115:194-199)

Key words: bronchoprovocation; chemical stability; histamine

Abbreviations: FDA = Food and Drug Administration; HDP = histamine diphosphate; HPLC = high-performance liquid chromatography

Extemporaneously prepared solutions of histamine acid phosphate (a.k.a. histamine diphosphate [HDP]) are used in inhalation tests to quantify airway responsiveness (*ie*, bronchoprovocation).¹ Some pulmonary function laboratories use them to confirm the diagnosis of asthma in a patient with a history of symptoms but normal results of pulmonary function tests on presentation. Also, they are used as a research tool to evaluate the long-term effects of various therapeutic interventions on airway responsiveness² and as a bioassay to determine the relative amount of β -agonist delivered to the lungs.^{3,4} Metha-

choline chloride has been used for these purposes and was often preferred over histamine because of the availability of a Food and Drug Administration (FDA)-approved commercial product (Provocholine). However, this product is no longer available in the United States, so both methacholine and histamine are equally inconvenient to obtain and require extemporaneous preparation by a pharmacist.

With these inhalation tests, the bronchoconstrictor response to a known concentration of histamine or methacholine is measured. Thus, the amount of bronchoconstrictor delivered to the airways is a major variable in interpreting the results. Obviously, the results of an inhalation test would be misleading if the test solutions contain significantly different amounts of bronchoconstrictor than what is indicated on the vial label. Such a circumstance would occur if the bronchoconstrictor solution was either inaccurately prepared or lost activity before use, *eg*, during storage.

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Manuscript received September 26, 1997; revision accepted May 12, 1998.

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There are several reports in the literature on the stability of histamine in solution.⁵⁻⁹ However, the results of these studies are difficult to interpret because of differences in test conditions or use of an assay that may not have differentiated histamine from its breakdown products (*ie*, not “stability-indicating”). For example, Nielsen et al⁵ concluded that solutions containing <0.5 mg/mL of HDP were stable for <4 weeks when stored at 20°C, while Rosenfeld et al⁶ concluded that a concentration of 0.125 mg/mL was stable for 3 months at room temperature. In addition, the authors of two other reports^{7,8} concluded that histamine solutions do not require protection from light during storage at 12°C, which is warmer than refrigerator temperature (4°C) and cooler than typical room temperature (20°C).

In preparation for submitting an investigational new drug application to the FDA for histamine bronchoprovocation, we developed a stability-indicating high-performance liquid chromatographic (HPLC) assay for HDP and tested solutions that were in use in our laboratory for a bronchoprovocation study. With this assay, histamine and its breakdown products are separated by the HPLC column and have different eluting peaks on the chromatogram. Therefore, breakdown products do not falsely contribute to the measured histamine concentration.

Our pharmacy had prepared two sets of solutions, each containing 100-mL multidose vials of 0.125 to 16 mg/mL HDP in carbonate buffered saline solution, which were stored in the refrigerator at 4°C with a 3-month expiration date. The procedures used for preparation and storage followed the recommendations of Juniper et al¹ except that the volume of the multidose vial was larger, the diluent was carbonate buffered saline solution instead of phosphate buffered saline solution, and sterilization was accomplished by cold filtration with a polymer (Millipore) filter instead of autoclaving. One of these sets was used several times a week for a bronchoprovocation study (set A), while the other was kept in the refrigerator as a backup and had never been used (set B). On study days, set A was removed from the refrigerator, used during a 6 to 8-h period, and then returned to the refrigerator. To our surprise, we found that a 0.5-mg/mL solution from set A actually contained only 0.23 mg/mL and a 0.125-mg/mL solution contained only 0.07 mg/mL. In contrast, the same dilutions from set B (never removed from the refrigerator) contained >90% of labeled amount of HDP.

Since it was unclear why the histamine degraded under the conditions of use in our laboratory (in and out of the refrigerator), we made several changes in the preparation and storage of these solutions. The diluent was changed to phosphate buffered saline

solution, the solutions were dispensed in 3-mL disposable syringes (unit-dose), and then they were immediately stored in the freezer. On the day of a histamine challenge, only one syringe of each concentration was removed from the freezer, and the set was allowed to thaw at room temperature while covered with a towel to prevent exposure to light.

The present investigation was conducted to determine the optimal storage conditions for histamine solutions used for bronchoprovocation. It was our hypothesis that intermittent exposure to room temperature and light caused degradation and that this could be prevented by storing the dilutions frozen in unit-dose syringes.

MATERIALS AND METHODS

Drug Substance

The histamine solutions used for this study were prepared from HDP hydrate powder (manufactured by Aldrich Chemical Company; Milwaukee, WI). Since the powder is a chemical and not a drug manufactured in an FDA-approved facility under “Good Manufacturing Procedures,” a sample was sent to a commercial testing laboratory (Custom Industrial Analysis Laboratories; St. Joseph, MO) to determine its content and purity. The laboratory performed all of the assays described in the *United States Pharmacopeia*, the compendial standard for drug products. The tests included identification of the substance, loss from drying, and the presence of arsenic, heavy metals, and organic volatile impurities. The histamine powder met all of the *United States Pharmacopeia* specifications and was free of contaminants.

Preparation of the Drug Product

The histamine solutions were prepared and packaged by the Pharmacy Department of Shands Hospital at the University of Florida. Briefly, the histamine powder was dissolved in phosphate buffered saline solution in amounts that gave the desired concentrations. The solutions were cold sterilized using a 0.22- μ m polymer (Millipore) filter and procedures were completed using aseptic techniques under a laminar flow hood. The final product was a unit-dose syringe containing 3 mL of one of the following concentrations of HDP in phosphate buffered saline solution: 0.125, 0.25, 0.5, 1.0, 2.0, 4.0, 8.0, and 16.0 mg/mL. The pH of all solutions was within the range of 6.7 to 6.8 U.

Microbiological Testing

Since bacterial contamination has been implicated in the degradation of histamine,^{10,11} five samples of each dilution were cultured immediately after preparation and after 9 months of storage in the freezer by the hospital microbiology laboratory. The samples were added to thioglycolate broth that will grow aerobic and anaerobic organisms, and incubated at 37°C for 72 h. The results of all cultures were negative.

HPLC Assay

A pump (Waters 510 Pump; Milford, MA) was used to deliver solvent at 1 mL/min to a 20-cm column (C18 uBondapak

Column; Waters). The mobile phase consisted of sodium dihydrogen phosphate 0.05 M adjusted to a pH of 2.5 with orthophosphoric acid, 0.004 M heptane sulfonic acid, and 1 mL t-butyl amine/L of mobile phase. All solvents used were of HPLC grade (Fisher; Fair Lawn, NJ). Eluting peaks were detected with a scanning detector (Spectra Physics Focus; San Jose, CA), which enabled UV spectral analysis between 200 and 400 nm and integration at 220 nm. Samples were injected by a 20- μ L fixed loop and quantitation was based on peak area determinations with a mean of three measurements.

Experimental Protocol

Effect of Intermittent Exposure to Room Light and Temperature: The first experiment was designed to confirm that the histamine solutions lost potency when intermittently exposed to light at room temperature as a result of repeated removal from the refrigerator. One syringe from each concentration (0.125, 0.25, 1.0, 2.0, 8.0, 16.0 mg/mL) was removed from a 4°C refrigerator, placed on a counter in a research laboratory (20°C), illuminated by fluorescent light for 6 h, and analyzed. The light intensity of the room was 375 foot-candles. After removing enough solution from the syringe to perform the analysis, the solutions were returned to the refrigerator. This procedure was repeated 2 and 4 days later.

Effect of Constant Exposure to Room Light at Room Temperature: In a second experiment to determine whether light exposure at room temperature caused the degradation, histamine dilutions of 0.125, 0.25, 0.5, and 1.0 mg/mL were left on a counter in the same laboratory (20°C) either wrapped in aluminum foil or left unprotected on the counter. The normal

fluorescent room lights (intensity, 375 foot-candles) were left on 24 h/d throughout the experiment. Each solution was analyzed daily in triplicate for 7 days.

Stability in the Refrigerator and Freezer: A final experiment was conducted to determine the effects of two typical storage conditions when syringes were protected from light. Histamine dilutions of 0.125, 0.25, 0.5, and 1.0 mg/mL were stored in the refrigerator (4°C) and freezer (-20°C) protected from light. Samples that were stored in the refrigerator were removed to obtain a 0.1-mL aliquot for analysis and immediately returned to the refrigerator. Samples were obtained and analyzed in triplicate weekly for 8 weeks. The frozen samples were removed from the freezer and placed in a refrigerator to thaw (approximately 1 h). A 0.1-mL aliquot of solution was removed from the syringe for analysis and then the syringe was returned to the freezer. The same syringes were analyzed in triplicate weekly for 4 weeks, then monthly up to 12 months.

RESULTS

Effect of Intermittent Exposure to Room Light and Temperature

After the first 6 h of exposure to room temperature and light, all dilutions contained approximately 100% of the concentration measured immediately after preparation (Fig 1, "Monday"). Thereafter, histamine concentrations progressively decreased after each successive 6-h exposure to room temperature

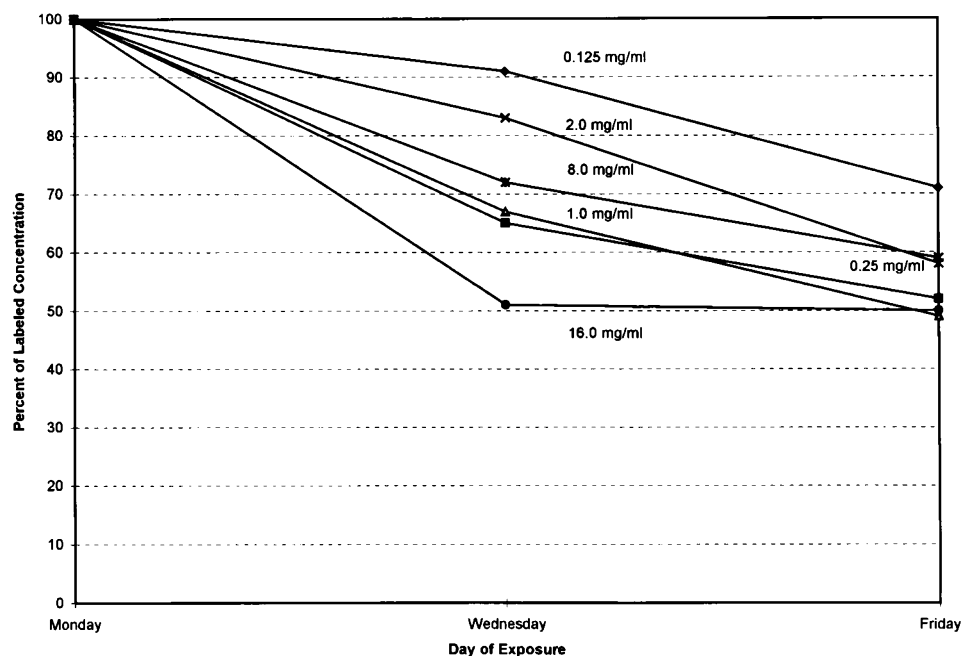


FIGURE 1. The effect of intermittent exposure to room light and temperature on six dilutions of histamine. One syringe from each concentration was removed from a 4°C refrigerator, placed on a counter in a research laboratory, illuminated by standard fluorescent lights for 6 h, analyzed, and returned to the refrigerator (Monday). This procedure was repeated 2 days later (Wednesday) and 4 days later (Friday). The solutions contained 0.125 (closed diamond), 0.25 (open square), 1.0 (open triangle), 2.0 (x), 8.0 (x), and 16.0 (closed circle) mg/mL of HDP. The solutions degraded with each successive 6-h exposure.

and light (Fig 1 “Wednesday”). The degradation was more apparent at higher concentrations, but the degree of breakdown was variable. After the third 6-h exposure on the fifth day of the same week, samples contained 49 to 71% of the initial concentration (Fig 1, “Friday”).

Effect of Constant Exposure to Room Light at Room Temperature

After 7 days of storage at room temperature, dilutions wrapped in aluminum foil contained 83 to 94% of the initial concentrations (Fig 2, “Dark”). In contrast, the same dilutions stored without protection from light contained only 20 to 37% of the initial concentration (Fig 2, “Light”). Most of the degradation occurred in the first 3 days of exposure to room light and temperature.

Stability in the Refrigerator and Freezer

There was no significant degradation of histamine in solutions stored in the refrigerator protected from light for 8 weeks. All samples retained about 90% of the initial concentration (data not shown). Dilutions stored in the freezer retained 90% of the initial potency 12 months after preparation (Fig 3). However, it appears that there was a slight degradation during the last month of storage.

DISCUSSION

The results of the present study clearly indicate that dilute solutions of HDP used for bronchoprovocation lose potency at room temperature when left on the counter unprotected from light either intermittently or continuously. Since the solutions remained sterile throughout each experiment, the degradation of histamine was not a result of bacterial contamination that could result in opening of the imidazole ring of histamine’s chemical structure.^{10,11} Thus, the most likely cause of degradation was exposure to the normal fluorescent light that illuminated the room at room temperature. Although a previous report indicated that ultraviolet light can cause degradation of histamine,⁹ two other reports indicated that histamine solutions were stable when exposed to direct fluorescent light at 12°C.^{7,8} In both of these studies, however, a colorimetric assay based on the Pauly reaction was used to measure histamine concentrations. It is possible that the histamine degradation may not have been detected by this assay, since it lacks the resolution, specificity, and sensitivity of a chromatographic method. In fact, McDonald et al¹² reported that they were unable to reproduce this colorimetric assay and relied on a liquid chromatography for their study of histamine stability after autoclaving. We have developed a

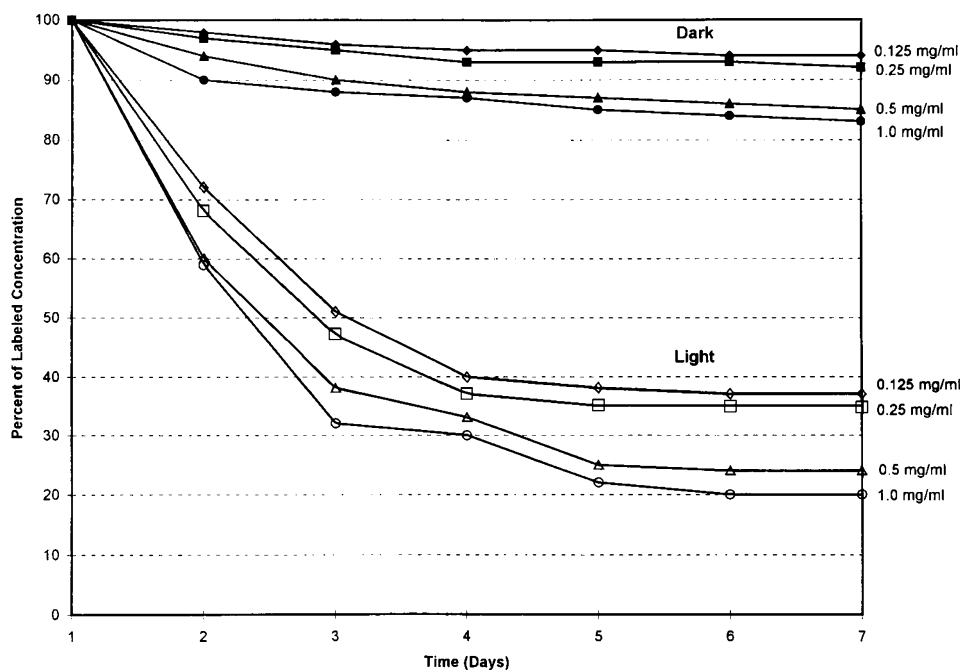


FIGURE 2. The effect of continuous exposure to room temperature with and without protection from light on dilute concentrations of HDP. The samples were placed on a counter in a research laboratory illuminated by fluorescent light either unprotected or wrapped in aluminum foil. These solutions were analyzed daily in triplicate for 7 days. Solutions exposed to light (open symbols) degraded much more rapidly than samples protected from light (filled symbols).

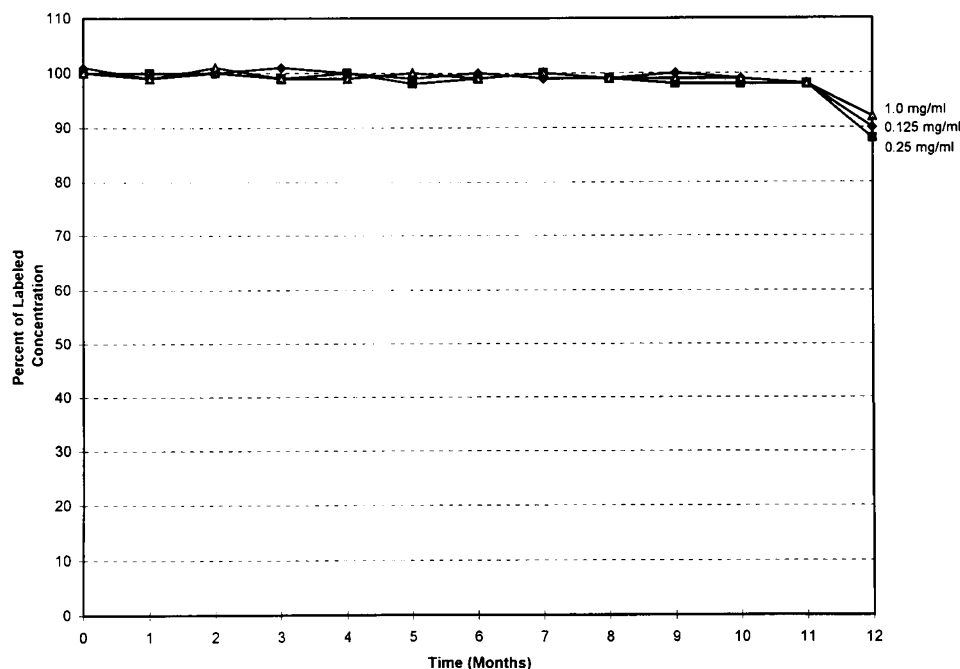


FIGURE 3. Stability of histamine solutions stored in the freezer and protected from light. Solutions were analyzed in triplicate weekly for 4 weeks, followed by monthly analysis up to 1 year. At 12 months, solutions contained approximately 90% of the labeled concentration. A small amount of degradation appeared to occur during the last month of storage.

stability indicating HPLC assay for histamine that separated the breakdown products from the parent compound. Breakdown products from solutions that lost potency could be identified at a different point on the chromatogram since they were eluted at a different time.

It is also likely that there is a temperature-light interaction in the degradation of histamine. Pratter et al⁷ and Marwaha and Johnson⁸ tested the effect of light at 12°C, whereas we evaluated the effects of light at 20°C. Since 12°C is neither room nor refrigerated temperature, the test conditions in those two studies do not seem to be clinically relevant. Also, we did not test the effects of light on solutions stored in the refrigerator or freezer since the light is turned off in most units when the door is closed. Such an experiment would be necessary to define the interaction of temperature and light on the degradation process, but it would not be clinically relevant to most refrigerators and freezers.

We found that all of the dilutions left on the counter top at room temperature without protection from light contained < 50% of the initial concentrations after only 3 days of continuous exposure. In contrast, Rosenfeld et al⁶ reported that similar concentrations maintained potency for 3 months at room temperature. Although they indicated that an HPLC assay was used for the analysis, the report was only published as an abstract so the details of their assay

and whether the samples were protected from light are not available for evaluation.

When we first discovered degradation of the multidose vials that we were using for histamine challenges, we asked our pharmacy to dispense each dilution in a 3-mL syringe so that we could freeze them until needed. In addition to providing a method of maintaining potency for a year, use of unit-dose syringes prevents bacterial contamination of the solutions and also prevents a dilution error during the challenge.

We conclude that histamine solutions used for bronchoprovocation lose potency if exposed to light at room temperature. We recommend that each dilution of histamine be stored in a unit-dose syringe in a dark refrigerator for 8 weeks or in a dark freezer for 12 months. When removed from the freezer or refrigerator, they should be used within 6 h and any unused dilutions should be discarded.

ACKNOWLEDGMENTS: The authors thank Rhonda Cooper, PharmD, Director of the Investigational Drug Service of the Pharmacy Department of Shands Hospital at the University of Florida, for meticulously preparing, labeling, and storing the histamine solutions. Also, we thank Laura Kennedy and Teresa Soard for word processing and carefully editing the typescript.

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